

ASSIGNMENTS

Class-2

Echelon Form, Row Reduced Form, Pivot Variables, Free variables

1. Find the rank of the following matrices

i) $\begin{bmatrix} 3 & -1 & 2 \\ 1 & 2 & -3 \\ 4 & 1 & -1 \end{bmatrix}$

ii) $\begin{bmatrix} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{bmatrix}$

iii) $\begin{bmatrix} 1 & 2 & 1 & -1 \\ 9 & 5 & 2 & 2 \\ 7 & 1 & 0 & 4 \end{bmatrix}$

iv) $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 0 \end{bmatrix}$

2. Find the special solutions to $Ax = 0$ by reducing A to its row reduced form R .

Where $A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix}$